Telecom Network Operations Assistant - Dataset Details

Dataset identity

Primary source: data/telecom_network_incidents_with_id.csv.

Observed rows: 100,000; observed columns: 28.

The dataset represents telecom network incidents/observations across performance, operations, load, signal, environment, security, geography, and fault-risk fields.

Schema and field groups

The canonical schema is files_load_context_files/schema.json and the normalized schema is files_load_context_files/schema.jsonl.

Context and geography: Incident_ID, Season, Weather, Issue Reported, City, State, Zip.

Performance: Cell Availability, Throughput, Latency, Packet Loss Rate, Call Drop Rate, Handover Success Rate, Jitter, Connection Setup Success Rate.

Operations: MTTR, Alarm Count, Critical Alarm Count, Parameter Changes, Successful Configuration Changes.

Load and environment: Data Usage, User Count, Signal Strength, Temperature, Humidity.

Security and target: Security Incidents, Authentication Failures, Fault Occurrence Rate.

Data quality and usage

The Synapt knowledge context describes the source dataset as 100,000 rows with 27 columns and no missing values; verify against the current CSV before production use.

Historical retrieval uses metadata/numeric filters plus semantic and lexical ranking; historical averages are descriptive and are not live telemetry.

The dataset contains synthetic/project data and should not be presented as a live operator network feed.

Embedding and retrieval

Local embedding model: BAAI/bge-small-en-v1.5 through sentence-transformers.

Vector store: ChromaDB; lexical retrieval: rank-bm25; orchestration combines semantic and lexical evidence.

The project also contains older commented OpenRouter embedding code; the active embedding path is local BGE.

Prediction features

Throughput prediction uses performance, operational, load, signal, environmental, security, and categorical context features.

Fault-risk prediction uses throughput, performance, operational, load, signal, environmental, and security features.

Models run only when the required live feature set is supplied; missing telemetry must not be silently imputed as current truth.